



Analog CMOS Circuit Design SS26

Practical Exercise I – Introduction to Cadence

Group

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Feedback

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Points **Mark**

Date **Signature**

No review without this standard form!



PE1

T1: DC Analysis, DC Sweep, Parameter Sweep

Hand Calculation

1. Calculate the signal transfer function

$$\frac{V_{out}}{V_{in}} = \frac{R_2}{R_1 + R_2}$$

- 2.

$$V_{out} = 0.5V_{in}$$

Vin (V)	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Vout (V)	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50

Vin(V)	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
Vout(V)	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	1.00

Vin(V)	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
Vout(V)	1.05	1.10	1.15	1.20	1.25	1.30	1.35	1.40	1.45	1.50

Vin(V)	3.1	3.2	3.3
Vout(V)	1.55	1.60	1.65

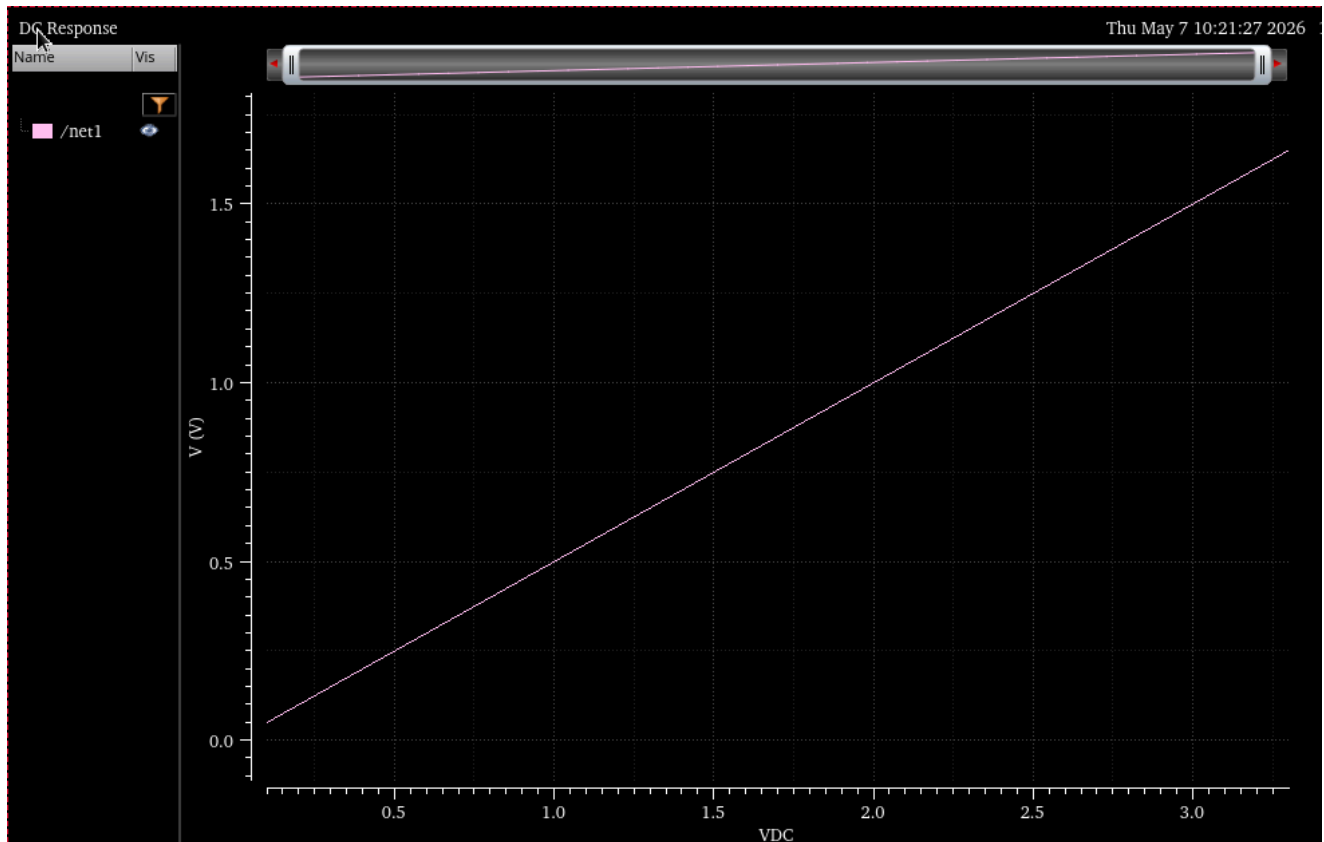
- 3.

$$V_{out} = 3.3V \frac{R_2}{1k\Omega + R_2}$$

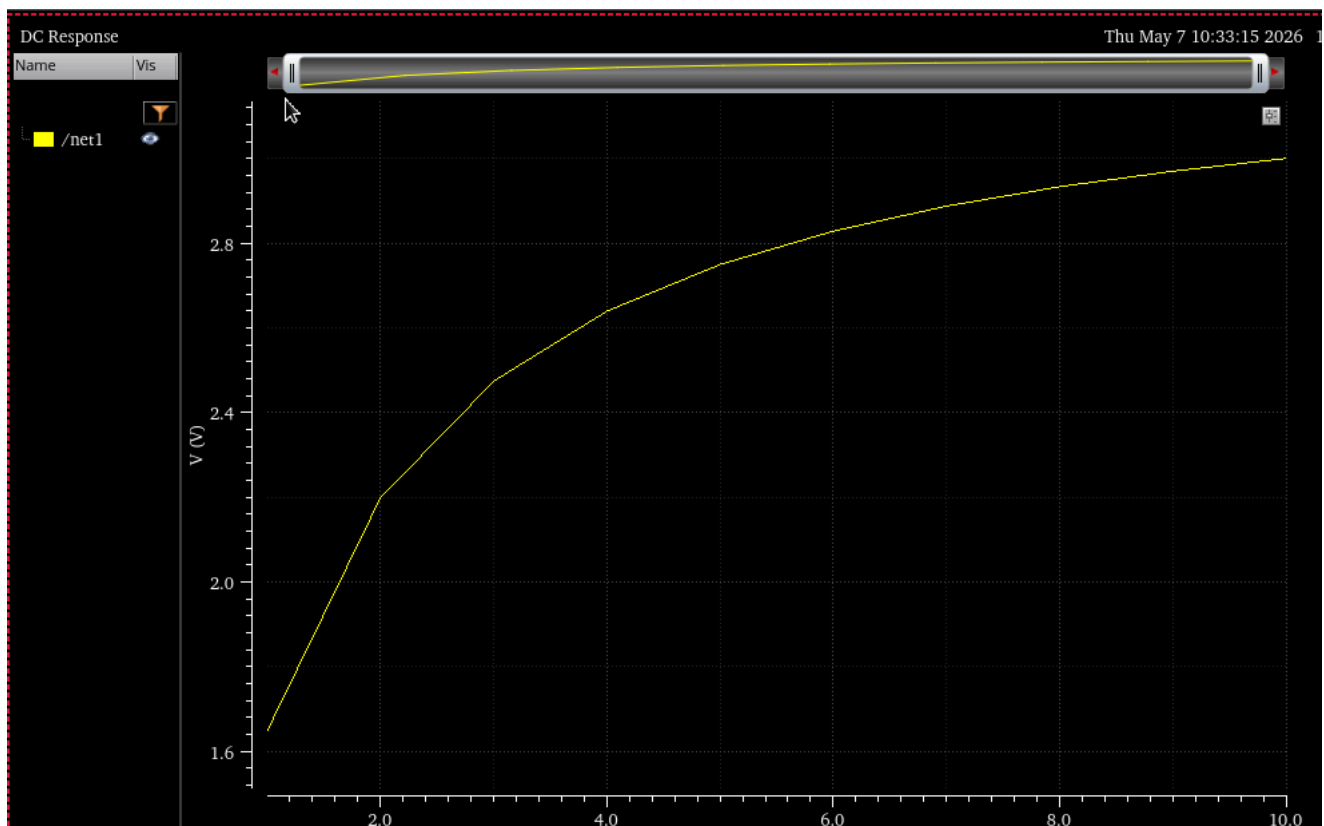
R2	1	2	3	4	5	6	7	8	9	10
Vout(V)	1.65	2.20	2.475	2.64	2.75	2.8286	2.8875	2.9333	2.97	3.0

Cadence Verify

b



c



T2: Transient Analysis, AC-Analysis

1. Transfer function

$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC} = \frac{1}{1 + j\omega RC}$$

Magnitude

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + (2\pi f RC)^2}}$$

Phase

$$\angle H(j\omega) = 0 - \arctan\left(\frac{\omega RC}{1}\right) = -\arctan(\omega RC)$$

2. -3db cut-off frequency

Cut-off frequency is defined as the frequency at which the magnitude of the circuit's frequency response drops to $1/\sqrt{2} \approx 0.707$ times its DC gain

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}}$$

$$\omega_c RC = 1 \implies \omega_c = \frac{1}{RC}$$

$$f_c = \frac{1}{2\pi RC}$$

$$RC = 12.5 \times 10^3 \times 8 \times 10^{-12} = 10^{-7} \text{ s}$$

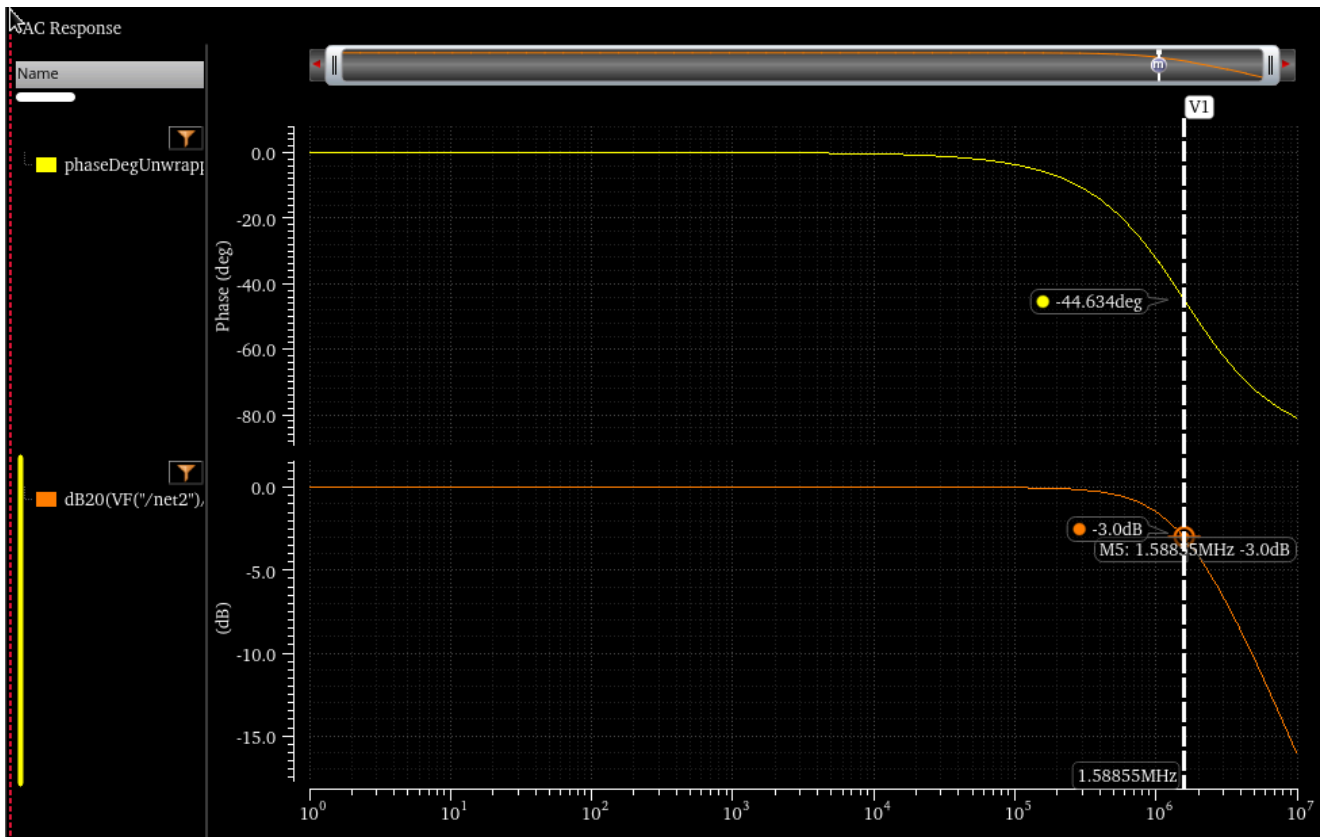
$$f_c = \frac{1}{2\pi \times 10^{-7}} \approx 1.59 \text{ MHz}$$

3. Sketch

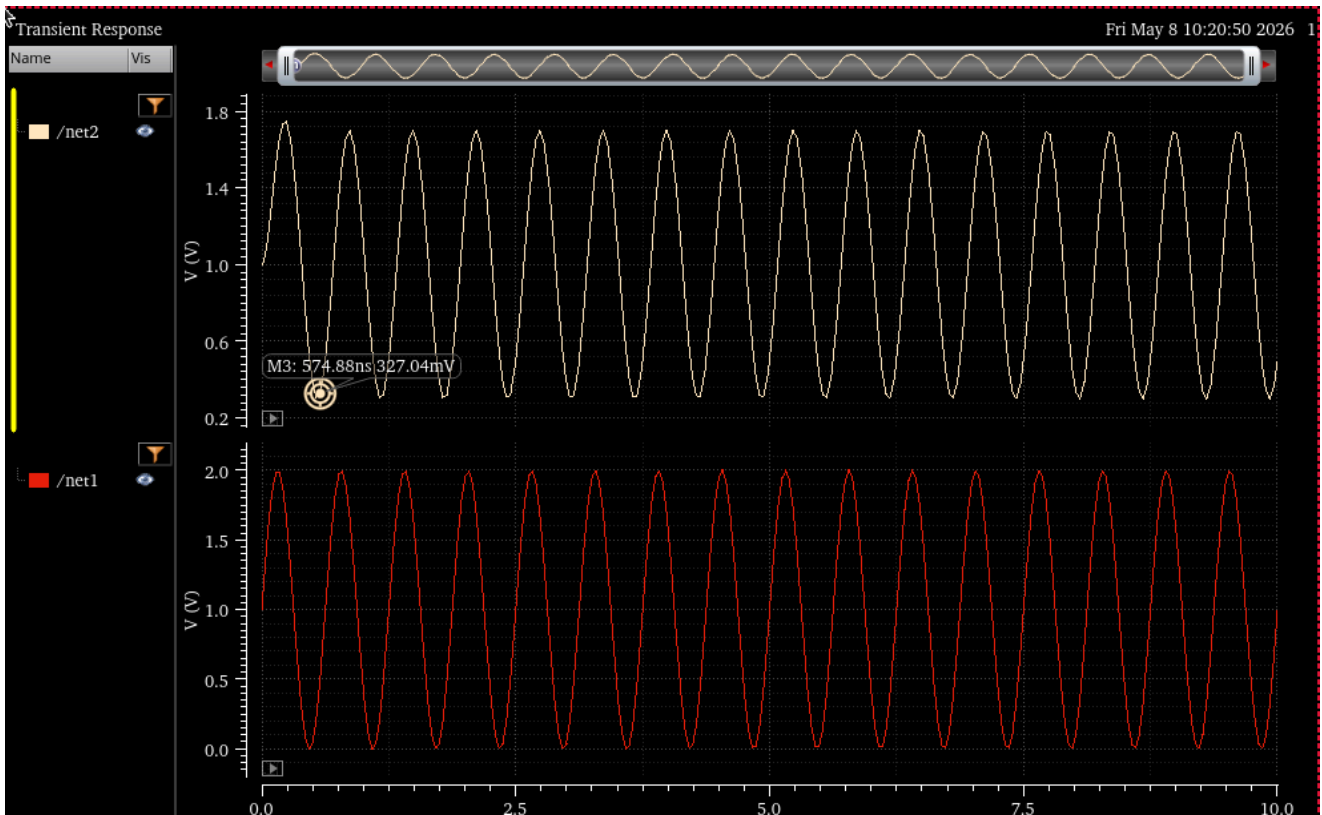
ac

$$f_c = 1.588 \text{ MHz}$$

$$\text{Phaseshift} \approx -45\text{deg}$$

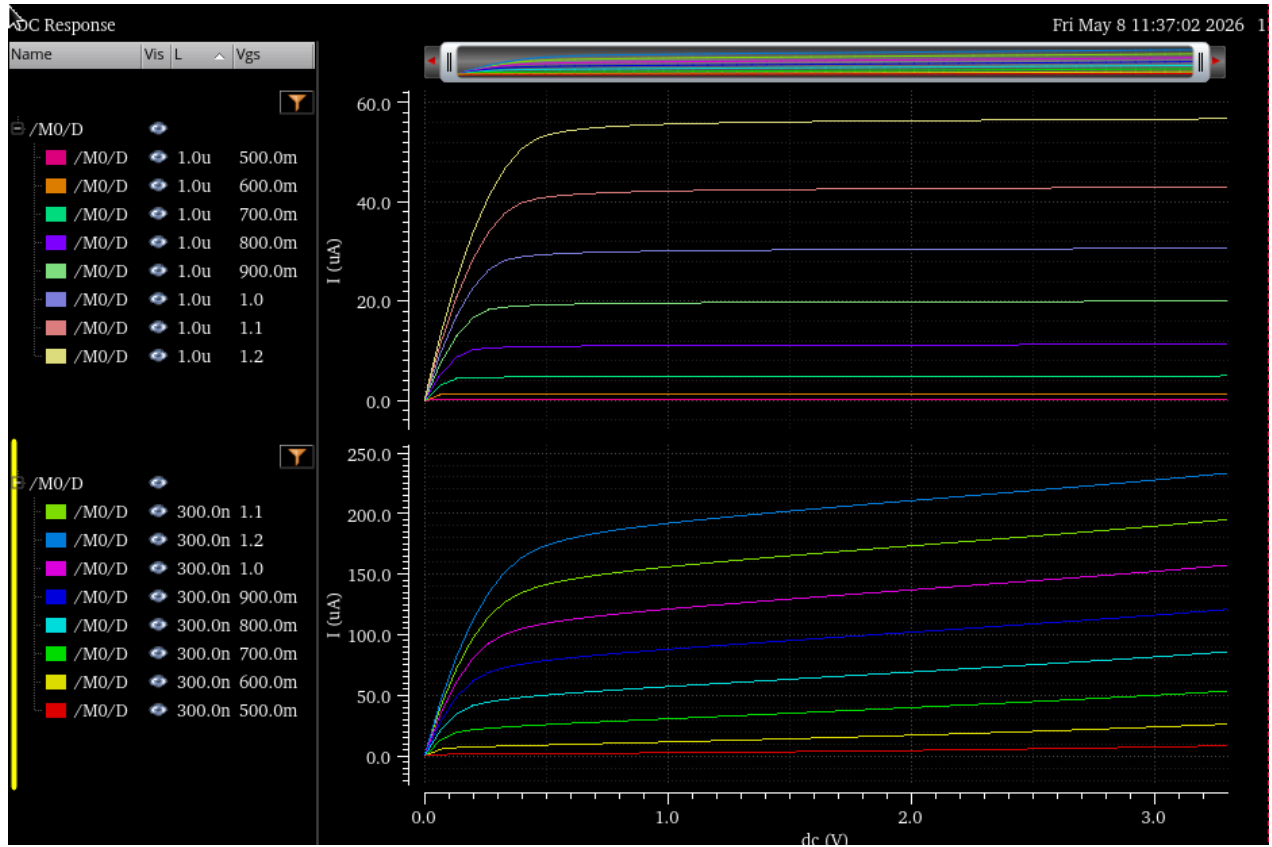


tran



T3: Transistor Transfer Characteristic Curve

1. Simulate the ID versus UDS characteristic



2.

$$\lambda = \frac{I_{D2} - I_{D1}}{I_{D1}V_{DS2} - I_{D2}V_{DS1}}$$

3.

- For short channel $L=0.35\mu\text{m}$, λ is higher because of stronger channel length modulation.
- The higher the V_{GS} , the bigger the λ .